

Linear Regression

C5025HF: concepts and intuition

Harper Adams University

What we're doing today

By the end, you should be able to:

- Explain what a *linear model* is (and what “linear” actually means)
- Fit and interpret a simple linear regression in R (`lm()`)
- Distinguish *prediction* vs *explanation* (and the traps in each)
- Check assumptions with diagnostic plots
- Report results in a professional, reproducible way
- Recognise when you need extensions: multiple predictors, interactions, blocks, random intercepts

Suggested pacing: - Part A: what a model is + regression idea - Part B: fit, interpret, confidence intervals, R^2 , p-values - Break: quick stretch - Part C: diagnostics + fixes - Part D: extensions + reporting + mini-assessment

Setup (R)

We'll use a small, friendly dataset.

If you prefer: swap in `mtcars` or your own agronomy dataset. We use penguins because it has clear relationships and is widely used in teaching.

Part A — What is a statistical model?

Definition: statistical model

A **statistical model** is a simplified mathematical description of a real system.

- It focuses on relationships between variables (and uncertainty),
 - not just *drawing a curve through points*.
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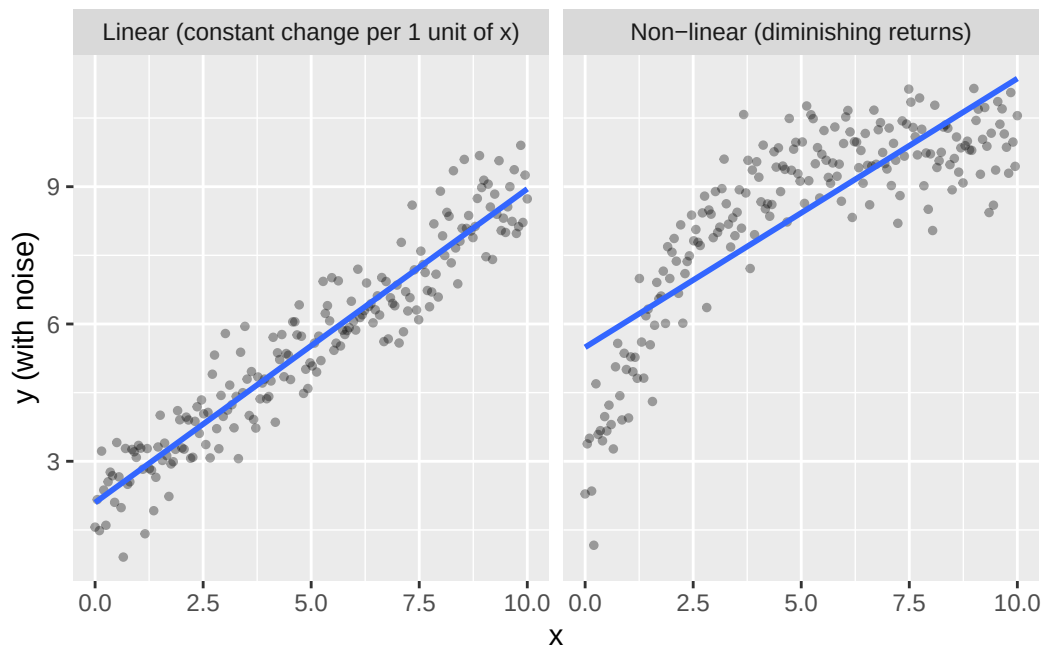
Linear vs non-linear (intuition)

A relationship can be:

- **linear additive**: a fixed change in x gives a fixed change in expected y
- **non-linear**: diminishing returns, saturation, thresholds, etc.

Use this to motivate “linear regression is a tool, not the truth”.

Linear vs non-linear



“Linear” means linear in the parameters

A **linear model** can include transformed predictors:

- $y = 0 + 1x + 2x^2 + \dots$ is *linear in* 's
- $y = 0 + 1\log(x) + \dots$ is *linear in* 's
- $y = 0 + \exp(1x) + \dots$ is *not* linear in 's

The linear model in matrix form

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

- $\mathbf{y}$: response/outcome
- $\mathbf{X}$: design matrix (predictors encoded as columns)
- $\boldsymbol{\beta}$: coefficients/parameters

- : residual error (unexplained part)
-

What regression is for

Simple regression often answers at least one of:

- **Prediction:** estimate y at new x
 - **Explanation:** quantify change in y per unit x
 - **Variance accounting:** how much variation in y is associated with x
 - **Inference:** hypothesis tests and confidence intervals
-

Regression to the mean (intuition)

Regression to the mean happens when you pick people (or things) because they had an **extreme result once**, then measure them again.

What you see: the extreme scores move closer to average on the second measurement.

Why? Every measurement has some **randomness** (luck, conditions, mood, error). When you pick “the best” or “the worst” from one measurement, you accidentally pick people who got lucky (or unlucky) that day. Next time, their luck is different—so their score drifts back toward their usual level.

Key examples:

- Students who bombed an exam → tend to do better on a retest (even with no extra study)
- “Rookie of the year” → often has a more normal second season
- Worst-performing stores → often improve next quarter (even with no intervention)

The trap: this looks like improvement (or decline), but it’s just randomness evening out. **Nothing actually changed.**

Why is it called “regression”?

The connection: linear regression literally models regression to the mean.

History: Francis Galton (1886) studied parent and child heights. He noticed:

- Very tall parents → children are tall, but closer to average height
- Very short parents → children are short, but closer to average height

He called this “**regression toward mediocrity**” (we now say “regression to the mean”).

Mathematics: When you fit a regression line $\hat{y} = \alpha + \beta x$, you’re estimating the **conditional mean**: the average y for a given x .

- If correlation is perfect ($\rho = 1$), the line has slope = 1 (no regression to mean)
- If correlation is imperfect ($\rho < 1$), the line is **flatter** than 45° , which means extreme x values predict y values closer to the mean

So: linear regression is the mathematical tool that quantifies how much extreme values regress back toward the mean. The phenomenon gave the method its name!

Caution: correlation causation

Regression describes association *in your dataset*.

To talk about causality you need:

- design (randomisation, controls),
- or a strong causal argument (assumptions you can defend).

Part B — Simple linear regression

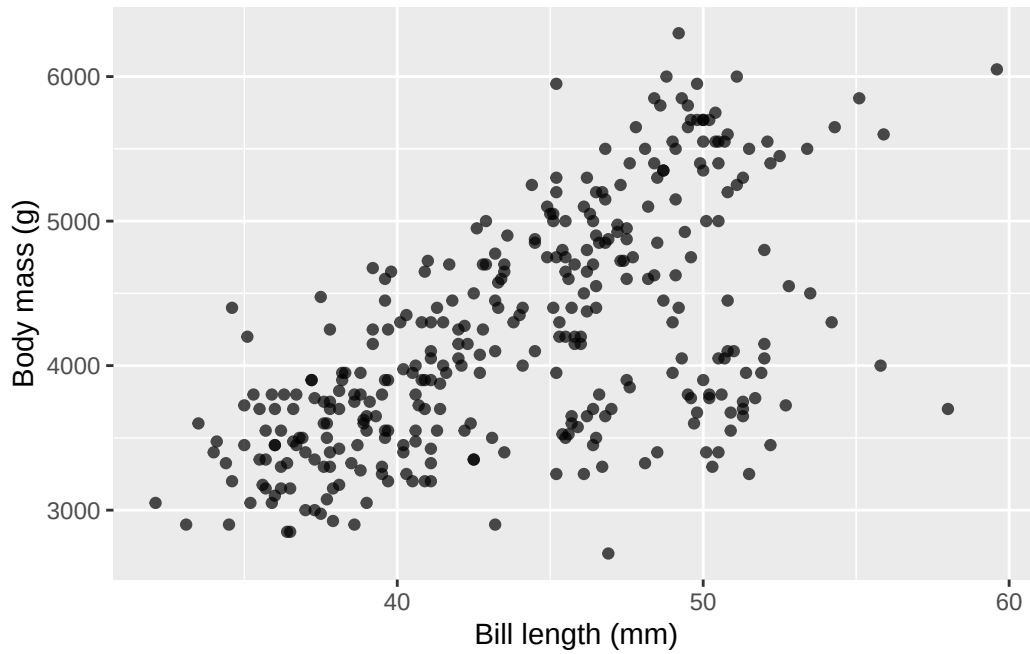
The simple regression model

$$y_i = \alpha + \beta x_i + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2)$$

- α : intercept (expected y when $x = 0$)
- β : slope (change in expected y for +1 in x)
- ε_i : residual (what’s left)

Example: body mass vs bill length

Here's the relationship we'll model:



Fit the model with `lm()`

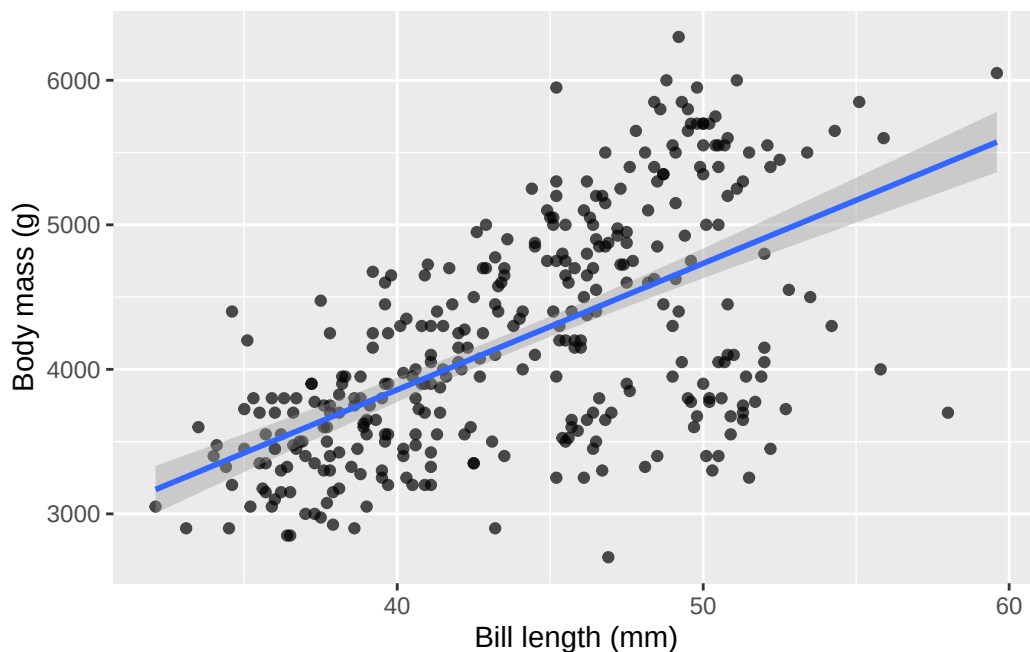
Call:

```
lm(formula = body_mass_g ~ bill_length_mm, data = penguins)
```

Coefficients:

(Intercept)	bill_length_mm
362.31	87.42

Add the regression line



Model coefficients

The fitted regression coefficients with 95% confidence intervals:

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	362.31	283.35	1.28	0.2	-195.02	919.64
bill_length_mm	87.42	6.40	13.65	0.0	74.82	100.01

Interpretation template:

- **Slope** : for a 1-unit increase in x , expected y changes by (holding other predictors constant)
- **Intercept** : expected y when $x = 0$ (sometimes not meaningful!)

Prediction vs residuals

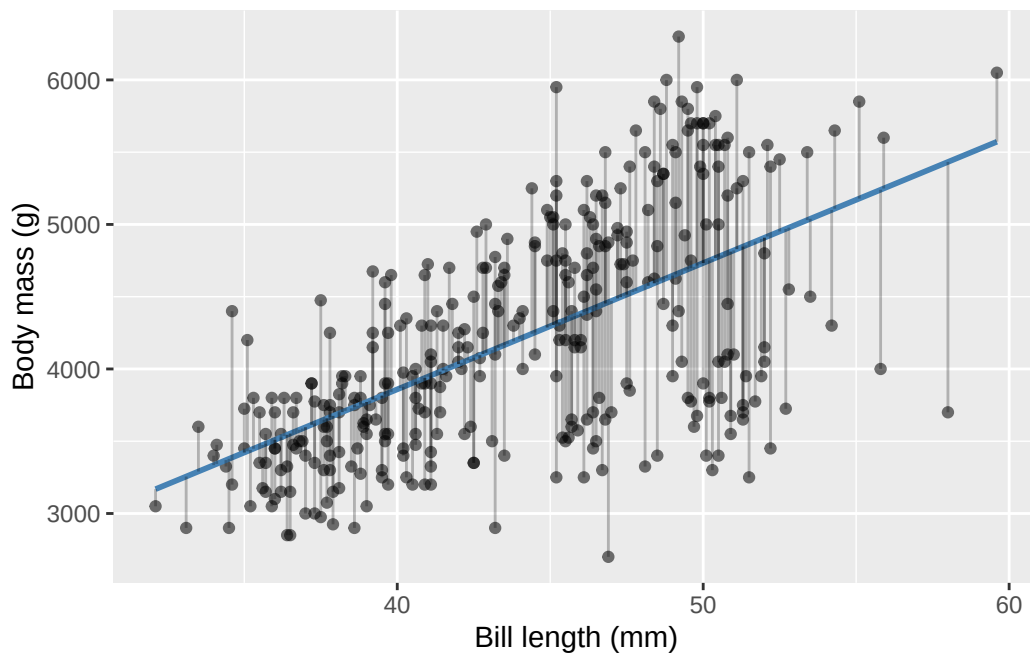
Two concepts:

- **Fitted value:** $\hat{y}_i$ (model's predicted mean)
- **Residual:** $e_i = y_i - \hat{y}_i$

First few observations with fitted values and residuals:

Body mass (g)	Bill length (mm)	Fitted	Residual
3750	39.1	3780.2	-30.2
3800	39.5	3815.2	-15.2
3250	40.3	3885.1	-635.1
3450	36.7	3570.4	-120.4
3650	39.3	3797.7	-147.7
3625	38.9	3762.8	-137.8

Residuals



Where do the coefficients come from?

Ordinary Least Squares (OLS)

Ordinary least squares chooses alpha, beta to minimise:

$$\sum_i (y_i - \hat{y}_i)^2$$

This is the **sum of squared residuals**.

Why *least squares*?

- Squaring makes all errors positive
- Larger errors are penalised more heavily
- The mathematics becomes tractable (closed-form solution)

Why *ordinary*?

Because we assume:

- all observations have **equal variance**
- all observations are **equally reliable**
- errors are **independent**

If these are not true, we move to:

- **Weighted least squares** (different variances)
- **Generalised least squares** (correlated errors)
- **Mixed models** (hierarchical structure)

So “ordinary” = *the simplest, default case*.

Ordinary least squares (OLS) chooses α, β that minimise:

$$\sum_i (y_i - \hat{y}_i)^2$$

This is why you’ll hear “*least squares line of best fit*”.

R² (variance explained)

How much variation does the model explain?

R ²	Adjusted R ²	Residual SD
0.354	0.352	645.433

Caution:

- high R² causal truth
- R² can increase just by adding predictors (use adjusted R², or better, validation)

p-values, confidence intervals, and power

Full model summary:

Call:

```
lm(formula = body_mass_g ~ bill_length_mm, data = penguins)
```

Residuals:

Min	1Q	Median	3Q	Max
-1762.08	-446.98	32.59	462.31	1636.86

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	362.307	283.345	1.279	0.202
bill_length_mm	87.415	6.402	13.654	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 645.4 on 340 degrees of freedom

Multiple R-squared: 0.3542, Adjusted R-squared: 0.3523

F-statistic: 186.4 on 1 and 340 DF, p-value: < 2.2e-16

What is being tested?

For each coefficient β_j , the null hypothesis is:

$$H_0 : \beta_j = 0$$

The test statistic is:

$$t = \frac{\hat{\beta}_j}{SE(\hat{\beta}_j)}$$

This follows a t-distribution under the model assumptions.

ANOVA view of regression

Analysis of Variance Table

Response: body_mass_g

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
bill_length_mm	1	77669072	77669072	186.44	< 2.2e-16 ***
Residuals	340	141638626	416584		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

ANOVA partitions variance into:

- **Model (Regression)** sum of squares
- **Residual (Error)** sum of squares

The F-statistic is:

$$F = \frac{MS_{model}}{MS_{error}}$$

If this ratio is large, the model explains much more variance than expected by chance.

Confidence intervals

	2.5 %	97.5 %
(Intercept)	-195.02364	919.6371
bill_length_mm	74.82279	100.0078

A 95% CI is the range of slopes compatible with the data under the model.

Interpreting the regression table

The coefficient table shows estimates, standard errors, t-statistics, and p-values:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	362.307	283.345	1.279	0.202
bill_length_mm	87.415	6.402	13.654	0.000

Typical output:

Term	Estimate	Std. Error	t value	Pr(>
(Intercept)	alpha	SE(alpha)	t	p
x	beta	SE(beta)	t	p

How to read this

- **Estimate:** the fitted coefficient (alpha or beta)
- **Std. Error:** uncertainty in that estimate
- **t value:** Estimate / Std. Error
- **Pr(>|t|):** p-value for H_0 : coefficient = 0

Intercept

The intercept is the expected value of y when **all predictors = 0**.

- Sometimes meaningful (e.g. baseline control)
- Often meaningless (e.g. bill length = 0 mm is impossible)

Slope (beta)

The slope is the **expected change in y per 1-unit increase in x**, holding other predictors constant.

Units always matter:

g per mm, kg per ha, t per °C, etc.

With factors

Call:

```
lm(formula = body_mass_g ~ species, data = penguins)
```

Residuals:

Min	1Q	Median	3Q	Max
-1126.02	-333.09	-33.09	316.91	1223.98

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3700.66	37.62	98.37	<2e-16 ***
speciesChinstrap	32.43	67.51	0.48	0.631
speciesGentoo	1375.35	56.15	24.50	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 462.3 on 339 degrees of freedom

Multiple R-squared: 0.6697, Adjusted R-squared: 0.6677

F-statistic: 343.6 on 2 and 339 DF, p-value: < 2.2e-16

Here:

- One level is the **reference** (baseline)
- Other coefficients are **differences from that reference**

Example interpretation:

- (Intercept) = mean of reference species
- speciesChinstrap = difference between Chinstrap and reference

With interactions

Model with interaction term:

Call:

```
lm(formula = body_mass_g ~ bill_length_mm * species, data = penguins)
```

Residuals:

Min	1Q	Median	3Q	Max
-918.76	-245.89	-8.65	238.44	1126.27

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	34.88	443.18	0.079	0.937
bill_length_mm	94.50	11.40	8.291	2.73e-15 ***
speciesChinstrap	811.26	799.81	1.014	0.311
speciesGentoo	-158.71	683.19	-0.232	0.816
bill_length_mm:speciesChinstrap	-35.38	17.75	-1.994	0.047 *
bill_length_mm:speciesGentoo	14.96	15.79	0.948	0.344

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 371.8 on 336 degrees of freedom

Multiple R-squared: 0.7882, Adjusted R-squared: 0.7851

F-statistic: 250.1 on 5 and 336 DF, p-value: < 2.2e-16

Interpretation:

- Main slope = effect in reference species
- Interaction term = **change in slope** for the other species
 - yield ~ variety * fertiliser: how different is the yield for variety A with fertiliser B from yield for variety A with fertiliser A?

The big rule

Every coefficient is a conditional comparison.

It is always interpreted **given the other terms in the model are held constant.**

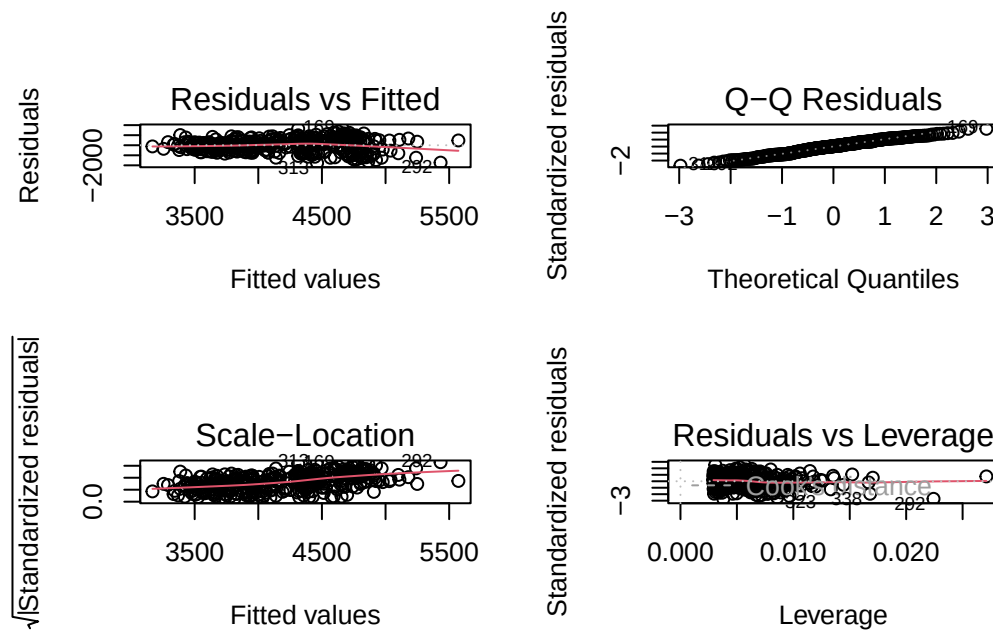
Part C — Assumptions and diagnostics

Core assumptions

In practice we check these assumptions about **residuals**:

1. Linearity of mean structure
2. Homoscedasticity (constant variance)
3. Independence
4. Approximate normality of residuals (mainly for small samples)

Diagnostic plots in R



What to look for:

- Residuals vs fitted: pattern? funnel shape?
- Q-Q: strong deviations?
- Scale-Location: heteroscedasticity
- Residuals vs leverage: influential points

Linearity: residual pattern

If you see curvature in residuals vs fitted:

- missing non-linearity (transform x, add polynomial term, or use a better model)

Analysis of Variance Table

Model 1: body_mass_g ~ bill_length_mm

Model 2: body_mass_g ~ poly(bill_length_mm, 2)

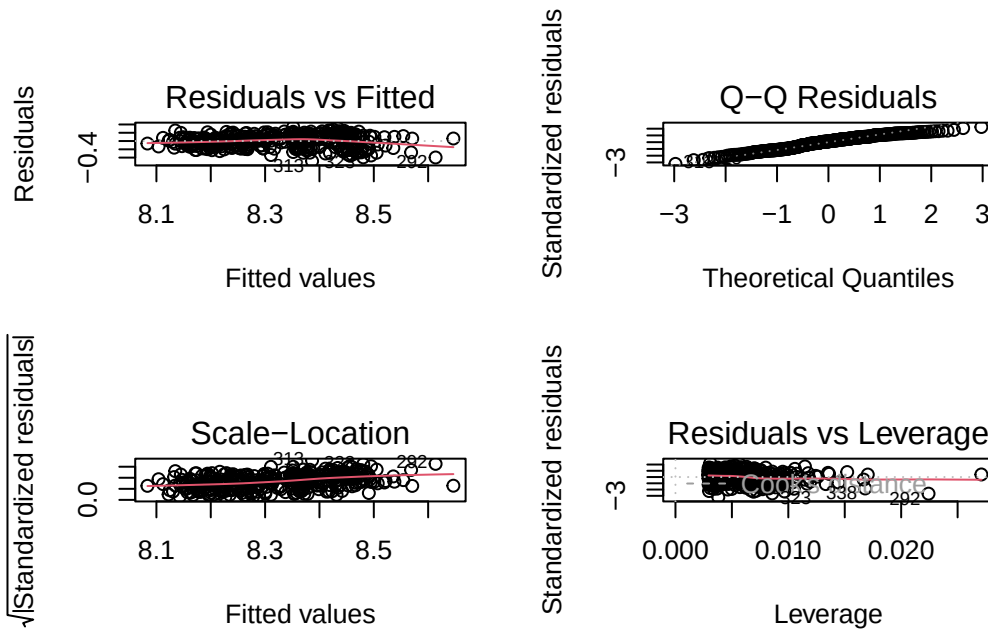
	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	340	141638626				
2	339	138722819	1	2915807	7.1254	0.007966 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Homoscedasticity: “fan” shape

If residual spread increases with fitted values:

- consider transforming the response (e.g., log)
- consider robust SEs (later modules), or different model family



Independence: design matters

Independence is usually **not** visible from a single plot.

Breaks happen with:

- repeated measures (same subject multiple times)
- spatial dependence
- time series

If independence is violated, standard errors can be wrong → misleading p-values.

Normality: mostly about small-sample inference

Normal residuals matter most when:

- sample size is small
- you need accurate confidence intervals/tests

With moderate/large samples, regression is often robust, but you still diagnose.

Influential points

Cook's distance identifies observations that strongly influence the fit. The top 10 most influential observations:

Body mass (g)	Bill length (mm)	Cook's D
3700	58.0	0.085
4000	55.8	0.032
3250	51.5	0.027
3450	52.2	0.026
3725	52.7	0.020
6300	49.2	0.018
3300	50.3	0.018
3400	50.5	0.017
3550	50.9	0.015
3400	50.1	0.015

Practical fixes (not “magic”)

When assumptions fail:

- **rethink the question** (is linear regression appropriate?)
- transform variables (log, sqrt)
- add terms (non-linear mean, interactions)
- model grouping (blocks/random intercepts)
- use alternative families (GLMs) or robust methods

Part D — Extensions you'll meet soon

Multiple regression (main effects)

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \varepsilon$$

Interpretation: **holding other predictors constant**.

```
# A tibble: 3 x 7
  term          estimate std.error statistic  p.value conf.low conf.high
<chr>         <dbl>    <dbl>    <dbl>   <dbl>   <dbl>   <dbl>
1 (Intercept)   -5737.    308.    -18.6  7.80e-54 -6343.   -5131.
2 bill_length_mm     6.05     5.18     1.17  2.44e- 1    -4.14    16.2
3 flipper_length_mm  48.1      2.01    23.9  7.56e-75    44.2    52.1
```

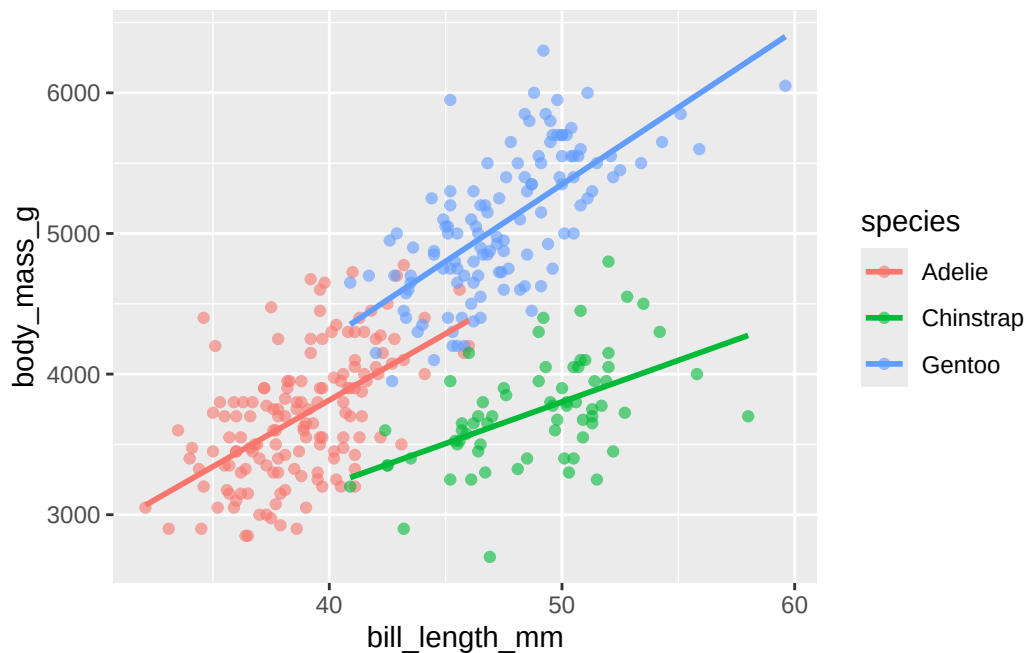
Interactions (effect modification)

An interaction means: the effect of x_1 depends on x_2 .

Model with interaction between bill length and species:

term	estimate	std.error	statistic	p.value
(Intercept)	34.88	443.18	0.08	0.94
bill_length_mm	94.50	11.40	8.29	0.00
speciesChinstrap	811.26	799.81	1.01	0.31
speciesGentoo	-158.71	683.19	-0.23	0.82
bill_length_mm:speciesChinstrap	-35.38	17.75	-1.99	0.05
bill_length_mm:speciesGentoo	14.96	15.79	0.95	0.34

Visualise:



Blocking (fixed effects for batches/sites)

Blocks are nuisance structure you *control for*.

Adding island as a blocking factor:

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	1225.79	243.18	5.04	0	747.45	1704.13
bill_length_mm	77.12	5.31	14.53	0	66.68	87.56
islandDream	-919.07	58.62	-15.68	0	-1034.38	-803.76
islandTorgersen	-523.29	85.55	-6.12	0	-691.57	-355.02

Random intercepts (preview)

If you have repeated measures, you often want:

$$y_{ij} = \beta_0 + b_{0i} + \beta_1 x_{ij} + \varepsilon_{ij}$$

- (b_{0i}) is a group-specific intercept deviation (random effect)

In R: `lme4::lmer()` (covered in mixed models).

Reporting results (what good looks like)

Minimum professional reporting:

- model formula and context (units!)
- coefficient estimate with CI (and p-value if required)
- a goodness-of-fit statement (R^2 , residual SD)
- diagnostics statement (what you checked, any issues)
- a plot (scatter + fitted line; residual checks)

Example sentence:

A linear model indicated that body mass increased with bill length ($\beta = \dots$ g/mm, 95% CI $\dots$), explaining $R^2 = \dots$ of variance; residual plots suggested $\dots$
